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Pose and camera calibration from cylinders via a silhouette based homotopy continuation

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Abstract: We address simultaneous camera calibration and pose estimation from multiple images of known 3D cylindrical objects taken by the same camera. Our approach introduces a tailored homotopy-based technique to handle non-linear constraints arising from the geometry of cylinder silhouettes. A novel correction step is introduced to recover improper solutions introduced by the homotopy. In contrast to existing methods, that address either calibration or pose estimation via algebraic approaches, our numerical solution addresses both tasks simultaneously achieving superior accuracy and robustness, as demonstrated through extensive experiments. Code is available at (Github).

Key-words: here, the keywords, of your thesis

1. Introduction

Camera calibration and localization are fundamental problems in Computer Vision, underpinning a wide range of applications such as 3D reconstruction, robotics, and extended reality. A variety of calibration and localization methods have been proposed, from traditional approaches based on point-wise 3D-2D correspondences [14, 17], to more recent techniques that exploit correspondences between higher-level geometric entities such as lines [7], conics [15], planes and quadrics [1, 16]. While point-based methods offer unparalleled accuracy in controlled settings, higher-order geometric primitives are more robust to noise and occlusions, and encode rich geometric constraints that can be leveraged during camera estimation.

In this work, we focus on camera calibration *and* localization from *cylinders*, that are common and easily detectable structures in many man-made environments, ranging from architectural elements (*e.g.*, pipes, handrails) to household objects (*e.g.*, bottles, cans, furniture legs) and infrastructure components (*e.g.*, street lamps, utility poles, tube lights). While calibration and pose estimation are traditionally treated as separate problems, we build upon the geometric constraints introduced in [6] to develop a novel formulation that estimates both the camera pose and the intrinsic calibration matrix. Under the assumption of constant intrinsics, this enables full camera resection directly from images of known 3D cylinders.

The core idea, illustrated in Figure 1, is to leverage the known 3D geometry to simplify the estimation process. The geometric constraints between the known cylinders and the unknown camera parameters form a system of nonlinear equations $\mathcal{S}_1(\mathbf{x}, \mathcal{L}_1)$. Instead of directly solving this complex system $\mathcal{S}_1$ with traditional algebraic methods like Gröbner basis or resultants, which are prone to numerical instability and scalability issues, we take a different approach. We generate a, *virtual system* $\mathcal{S}_0(\mathbf{x}, \mathcal{L}_0)$ designed around the very same 3D scene and a configuration of synthetic cameras with random but known parameters. By design, we know the solution

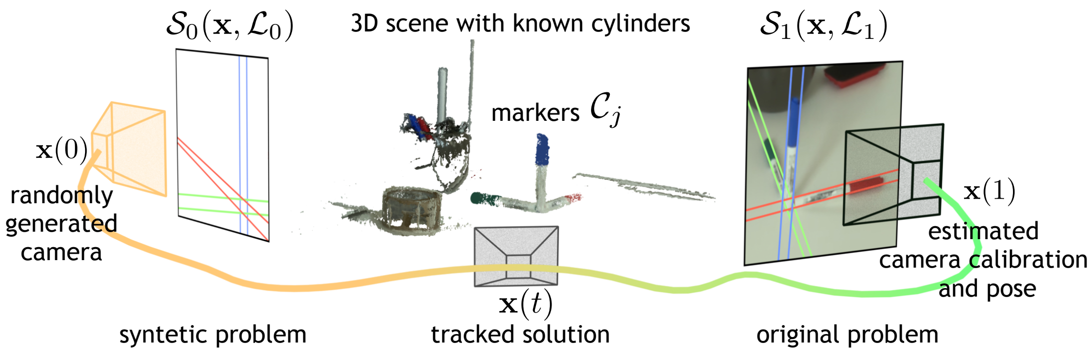


Figure 1: Our method for $N = 1$ camera and $M = 3$ cylindrical objects (3 markers). The input is a known 3D scenes and images with estimated silhouette lines. The output is the calibration and the pose of the cameras that acquired the images. Rather than tackling the problem on the input data, we move to a simpler random problem with a known solution $\mathbf{x}(0)$. Our silhouette-based parametric homotopy allows to track $\mathbf{x}(0)$ to the solution $\mathbf{x}(1)$ of the original problem. Our formulation support multiple cylinders and multiple images.

$\mathbf{x}(0)$ for $\mathcal{S}_0$, since it corresponds to the parameters we used to generate the synthetic cameras. Then, we continuously track the known solution $\mathbf{x}(0)$ as we transform the problem towards the original, system $\mathcal{S}_1$, using a tailored homotopy continuation parametrized in terms of silhouette lines. At the end, we obtain the desired calibration and pose $\mathbf{x}(1)$. By using this approach, we influence the trajectory of the solutions, guiding the solver toward more plausible configurations. We couple our solver with a dedicated post-processing step designed to discard spurious solutions that do not correspond to physically valid camera configurations. Compared to existing methods, our approach is not only more general – being capable of estimating both camera calibration and pose, unlike alternatives that address only one of the two task – but also achieves superior accuracy and improved robustness to noise, as demonstrated by our experiments.
All in all, our contributions can be summarized as follows:

- We present a *unified solution* for calibration and pose estimation, unlike existing approaches that typically address these two problems separately.
- We conceive a *novel solver*, that, exploiting a silhouette-based parametrization, simplifies the original problem into a one with a known solution, which is then lifted via homotopic continuation to the original one.
- We propose a geometric *correction step* to deal with unavoidable spurious solutions.
- Our method is *more accurate* and *robust* to noise compared to alternative solutions.

2. Problem formulation

We consider a 3D scene containing M cylinders $\mathcal{C}_j$ in general position, each with known radius, position, and axis direction $\mathbf{w}_j$. A camera with unknown but fixed intrinsic matrix K captures N images $\{I_1, \dots, I_N\}$ of the scene. Each image I_i is associated with an unknown camera matrix $\mathsf{P}_i = \mathsf{K}[\mathsf{R}_i \,|\, \mathbf{t}_i]$, where R_i and $\mathbf{t}_i$ denote the rotation and translation of the camera in view i. In each view I_i, each cylinder $\mathcal{C}_j$ projects onto a pair of *contour lines*, denoted l_{ij}^h with $h = 1, 2$ (see Figure 2 that displays one of the two silhouette lines of $\mathcal{C}_j$). These lines correspond to the visible contours of the cylindrical surface. Geometrically, each l_{ij}^h arises from the intersection between the image plane and the planes Π_{ij}^h that pass through the optical center of the camera P_i and are tangent to the surface of $\mathcal{C}_j$.
Given in input the parameters of the 3D cylinders $\{\mathcal{C}_j\}_{j=1}^M$ and a collection of silhouette lines $\{l_{ij}^h\}$ observed in the images, our aim is twofold:
1. Camera calibration: estimate the intrinisc parameters of the calibration matrix

$$\mathsf{K} = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}, \tag{1}$$

that correspond to the focal lengths f_x, f_y, the skew s, and the coordinates of the principal point (c_x, c_y). Depending on the assumptions made on the intrinsic parameters, different variants of the problem can be considered. By introducing suitable linear constraints – such as zero skew ($s = 0$), known aspect ratio ($f_x = \alpha f_y$), or a centered principal point – the original set of five unknowns can be reduced.
2. Pose estimation: estimate the extrinsic parameters, namely rotations R_i and translations $\mathbf{t}_i$, for each image I_i.

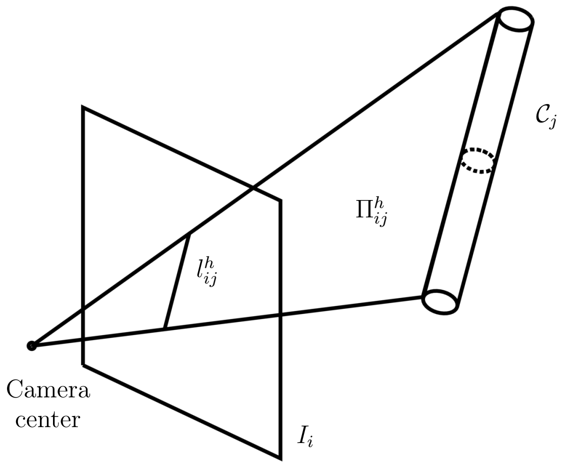


Figure 2: The geometric constraints between a cylinder $\mathcal{C}_j$ and one of its silhouette lines l_{ij}^h. The backprojection of l_{ij}^h is a plane Π_{ij}^h tangent to C_{ij} (3) and passing to the point at infinity of the cylinder(7), *i.e.*, parallel to its axis.

	Guo[7]	Ding [3]	Gummerson [4, 5]	Ours
Calibration	f ✗	K ✓	f ✗	K ✓
Pose estimation	R ✗	✗	R, **t** ✓	R, **t** ✓

Table 1: Comparison of camera estimation methods and their outputs for calibration and pose estimation. (✗: not addressed, ✗: partial output, ✓: fully solved). Our method uniquely solves both tasks and provides all parameters.

3. Related work

Several methods have addressed the estimation of camera intrinsics and pose from geometric primitives such as lines and conics. *Line-based approaches* include early works as [8], which jointly estimate K and 3D structure from multi-view line correspondences. Interestingly, Guo et al. [7] simplify the problem by assuming a known camera center and using two vanishing points to estimate the rotation R and focal length f. While the input of their method is similar to our, as the two 3D vanishing points can be identified with the two cylinder axis direction, their approach does not recover the translation, $\mathbf{t}$ nor the full set of intrinsic parameters. *Conic-based methods* exploit the relation between conic sections – typically circles – on planar calibration targets, and their projections on image planes to improve the accuracy and robustness of corner-based calibration procedure (*e.g.*,[13]).

Our focus is on the less explored class of *quadric-based* approaches that leverage the multi-view constraint provided by 3D quadrics as calibration and localization cues. We consider methods based on degenerate quadrics such as cylinders, whose surfaces project to line silhouettes under perspective projection. Gummeson et al. [4] propose minimal solvers for pose estimation using line silhouettes, but assume known intrinsics. Interestingly, under some circumstances, they show that they are also able to recover the focal f. Their more recent work [5] focuses on relative pose from silhouettes across two views, again assuming fixed intrinsics that must be known in advance. Our approach is directly inspired by these methods but goes further by addressing the full calibration problem. Moreover, rather than rely on Grobner basis, we exploits the advantages of homotopy continuation to geometrically inform the solution process. A further relevant contribution is the work by Larsson et al. [9], which investigates the multi-view geometry of mutually parallel cylinders. Their formulation allows a structure-from-motion pipeline based on silhouette lines, under the strong assumption that all cylinders share a common axis direction $\mathbf{w}_j = \mathbf{w}_k$ for all $j, k = 1, \ldots, M$. The emphasis is on recovering $\mathsf{R}_i, \mathbf{t}_i$ and the parameters of the cylinders $\mathcal{C}_j$ up to scale, assuming known intrinsics. In contrast, our method handles arbitrary cylinder orientations, assume the cylinders to be known and jointly estimates both intrinsic and extrinsic parameters, enabling complete camera resection. Cylinder sections are exploited by Ding et al. in [3]. This method leverages the known geometry of the target—specifically, the circular end-faces of the cylinder—and formulates constraints based on their elliptical projections. Table 1 summarizes the most relevant methods in terms of their output. Only our approach estimates both the full camera calibration K and the camera poses $\mathsf{R}_i, \mathbf{t}_i$.

4. Geometric constraints on cones and silhouette lines

Following [4, 5], we briefly recall some geometric properties of cylinders and their image projections, which form the basis for the constraints in our systems. A cylinder $\mathcal{C}$ can be interpreted as a cone with its vertex $\mathbf{V}$ lying on the plane at infinity. Algebraically, both cones and cylinders are represented by a singular quadric surface with matrix Q such that a point $\mathbf{X}$ belongs to $\mathcal{C}$ if and only if $\mathbf{X}^\top \mathsf{Q}\mathbf{X} = 0$.
A cylinder is a generate quadric and as such the rank of it's matrix Q is not full. The vertex $\mathbf{V}$ lies in the null-space of Q, that is, $\mathsf{Q}\mathbf{V} = \mathbf{0}$. For cylinders, this vertex takes the form $\mathbf{V} = (\mathbf{w}, 0)^\top$ in homogeneous coordinates, where $\mathbf{w}$ defines the direction of the cylinder's axis.

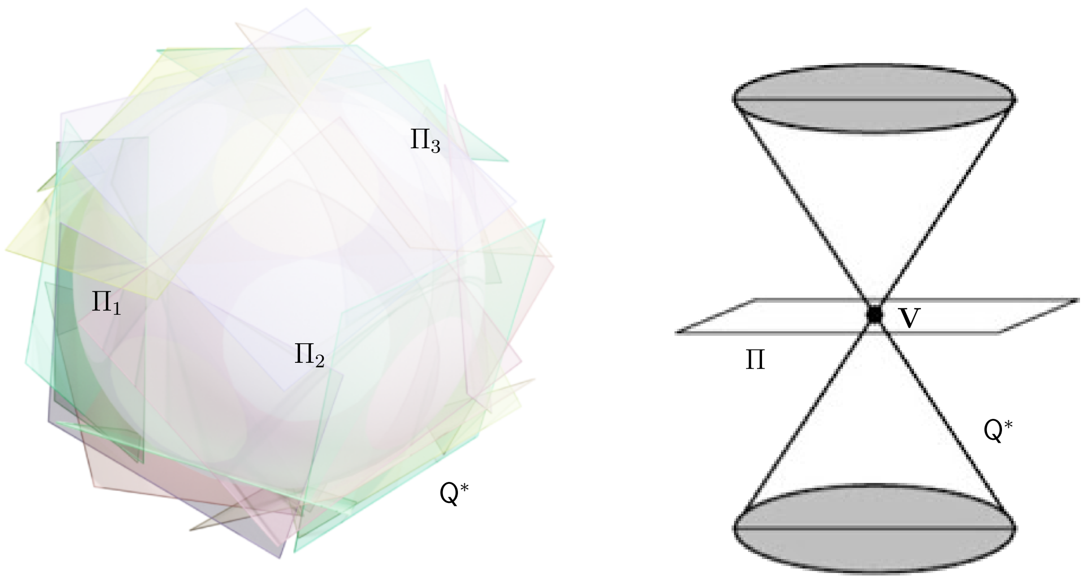
Rather than working directly with Q, it is convenient to use the dual quadric $\mathsf{Q}^* = \mathsf{Q}^{-1}$, which encodes the set of all planes tangent to $\mathcal{C}$. See Figure 3a
Technically speaking, all planes containing $\mathbf{V}$ are tangent to Q but not all of them uniquely define the surface Q^*. See Figure 3b. To uniquely identify the dual quadric Q^* in terms of the tangent planes two conditions must be satisfied:

$$\begin{cases} \Pi^\top \mathsf{Q}^* \Pi = 0, \\ \Pi^\top \mathbf{V} = 0. \end{cases} \tag{2}$$

These two conditions can be formulated in terms of silhouette lines l^h. The backprojection of each silhouette line, given by $\mathsf{P}^\top l^h$, defines a 3D plane that is tangent to the cylinder $\mathcal{C}$ and passes through the camera center. Moreover, since this tangent plane is aligned with the axis of the cylinder, $\mathsf{P}^\top l^h$ must contain the point at infinity $\mathbf{V}$ corresponding to the direction of the cylinder's axis. As a result, each visible silhouette line l^h induces two constraints on the camera parameters of the form:

$$l^{h\top} \mathsf{P}\mathsf{Q}^* \mathsf{P}^\top l^h = 0 \tag{3}$$

$$l^{h\top} \mathsf{P}\mathbf{V} = 0. \tag{4}$$



(a) The plane envelop of a dual quadric Q^*. Tangent planes Π_i define the overall surface

(b) An example of a plane Π tangent to a cone $\mathcal{C}$, which does not define the quadric surface Q^*

5. Homotopy Continuation

Homotopy continuation is a powerful numerical–symbolic method for solving systems of nonlinear (especially polynomial) equations by deforming a solvable start system into the target system and tracking solution paths continuously.
More formally, given $F(x) = 0$ to solve where $x \in \mathbb{C}^n$, we construct a homotopy

$$H(x,t) = (1-t)\,G(x) + t\,F(x), \tag{5}$$

$$H(x,0) = G(x), \;\; H(x,1) = F(x), \tag{6}$$

where G has known regular solutions.
Starting from a known solution $x(0)$, we numerically follow the curve $x(t)$ satisfying $H(x(t),t)=0$. Writing

$$\frac{\partial H}{\partial x}\dot{x}+\frac{\partial H}{\partial t}=0 \quad \Rightarrow \quad \dot{x}=-\left(\frac{\partial H}{\partial x}\right)^{-1}\frac{\partial H}{\partial t},$$

we use predictor steps (e.g. Euler, secant, Hermite) and corrector steps (Newton refinement). See Algorithm 1, Figure 3

Algorithm 1 Homotopy Path Tracking via Predictor-Corrector

Require: Initial solution x_0, initial time $t_0=0$, step size Δt

$k \leftarrow 0$
$x_k \leftarrow x_0$, $t_k \leftarrow t_0$
while $t_k < 1$ **do**
  Compute $\dot{x}_k = -\left(\frac{\partial H}{\partial x}(x_k,t_k)\right)^{-1}\frac{\partial H}{\partial t}(x_k,t_k)$
  $\tilde{x} \leftarrow x_k + \Delta t\, \dot{x}_k$ (Predictor: Euler or higher order)
  $t_{k+1} \leftarrow t_k + \Delta t$
  $x_{k+1} \leftarrow$ Newton's method to solve $H(x,t_{k+1})=0$ starting from $\tilde{x}$
  $k \leftarrow k+1$
end while
return x_k (approximate solution to $F(x)=0$)

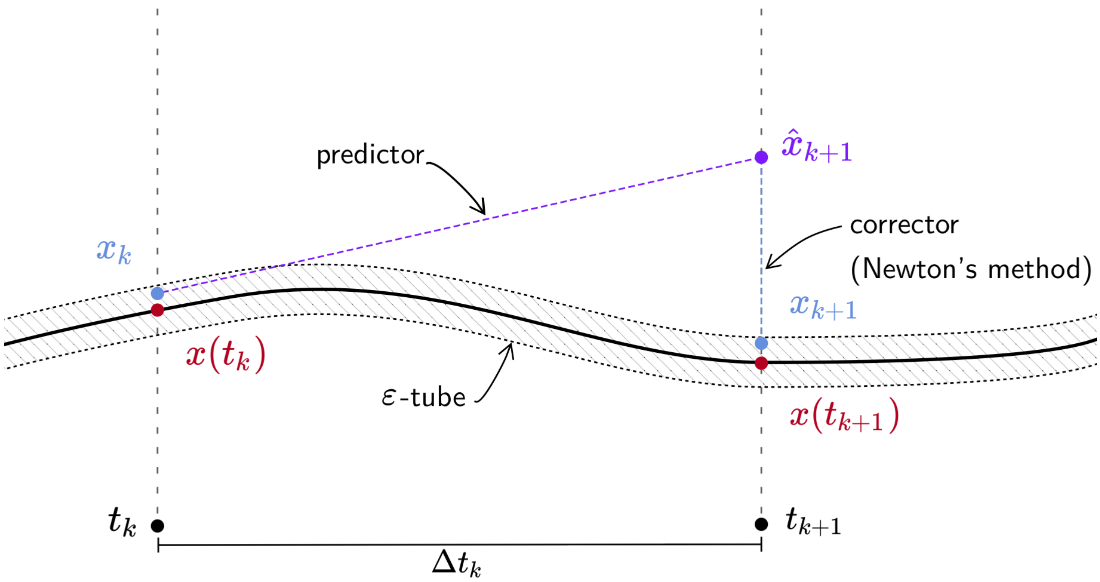


Figure 3: A sketch of the predictor-corrector method. We compute a prediction $\tilde{x}=x_k+\Delta t\,\dot{x}_k$ using any ODE method and then correct using Newton's method.

6. Method

By collecting the pairs of geometric constraints associated with each visible silhouette line l_{ij}^h across all images, we obtain a system of polynomial equations of the form (3) and (4). We refer to this as the *original system* $\mathcal{S}_1$, as it encodes the multi-view geometric relationships between the known 3D scene and the actual, but unknown, camera matrices $\mathsf{P}_i=\mathsf{K}[\mathsf{R}_i,\mathbf{t}_i]$. Solving $\mathcal{S}_1$ directly is challenging due to its degree, multiple solutions, and the presence of non-linearities. To overcome these challenges, we design a custom homotopy based on silhouette lines. We begin by constructing a *synthetic*, geometrically consistent starting system $\mathcal{S}_0$, which collects the constraints between the 3D cylinders and a set of silhouette lines generated by known synthetic cameras. As sketched in Figure 1, at high level, we continuously deform $\mathcal{S}_0$ into the target real system $\mathcal{S}_1$ through a homotopy parametrized by $t\in[0,1]$, gradually transforming the synthetic silhouette lines into the observed ones in a geometric consistent way, while tracking the corresponding camera parameters, denoted by $\mathbf{x}(t)$, throughout the process. At $t = 0$, the solution $\mathbf{x}(0)$ of $\mathcal{S}_0$ corresponds to the known synthetic cameras, whereas the tracked solution at $\mathbf{x}(1)$ yields the estimated cameras configuration that solves the real original problem.

6.1. The original target system $\mathcal{S}_1$

Let $\mathcal{L}_1$ denote the collection of all visible silhouette lines from all cylinders across all images. Each visible silhouette line $l^h_{ij} \in \mathcal{L}_1$ contributes two quartic equations, given by (3) and (4), which together define the original system $\mathcal{S}_1$. With a slight abuse of notation, since the set of silhouette lines $\mathcal{L}_1$ defines the parameters of the system, we can write $\mathcal{S}_1(\mathbf{x}, \mathcal{L}_1)$ to emphasize this dependency. As observed in [5], (4) involves only the intrinsic K and the rotation R. For a cylinder $\mathcal{C}_j$ captured in the i-th image I_i by a camera $\mathsf{P}_i = \mathsf{K}[\mathsf{R}_i, \mathbf{t}_i]$ we have

$$l^{h\top}_{ij}\mathsf{K}[\mathsf{R}_i|\mathbf{t}_i]\begin{pmatrix}\mathbf{w}_j\\0\end{pmatrix} = l^{h\top}_{ij}\mathsf{K}\mathsf{R}\mathbf{w}_j = 0. \tag{7}$$

Thus, one could solve for K and all R_i from all the equations in $\mathcal{S}_1$ of the form (4). Then from all the equations of the form (3), solve for the translations $\mathbf{t}_i$.

Using this approach, assuming V visible silhouette lines in total, the system $\mathcal{S}_1$ contains V equations of the type (4). The number of unknowns is $3N + 5$, corresponding to the 3 rotation parameters per view and the 5 intrinsic parameters of the camera. Therefore, solving for the camera intrinsics and all rotations requires satisfying the inequality:

$$V \geq 3N + 5. \tag{8}$$

Once K and the rotations R_i are estimated, the translations $\mathbf{t}_i$ can be recovered from the remaining V equations of type (3), which are quadratic in $\mathbf{t}_i$. In practice, given M cylinders, under the assumption of full visibility (*i.e.*, $V = 2MN$ silhouette lines), $\mathcal{S}_1$ can be solved for the pair (M, N) marked with in Table 2. A single cylinder $M = 1$ does not allow for solution, while beyond $M = 6$ or $N = 9$, $\mathcal{S}_1$ is always solvable.

M / N	1	2	3	4	5	6	7	8	9
2	✗	✗	✗	✗	✗	✗	✗	✗	✓
3	✗	✗	✓	✓	✓	✓	✓	✓	✓
4	✗	✓	✓	✓	✓	✓	✓	✓	✓
5	✗	✓	✓	✓	✓	✓	✓	✓	✓
6	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 2: Assuming all the solhouette lines are visible, $\mathcal{S}_1$ can be solved when the number of cylinders M and the number of view N satisfy $2MN > 3N + 5$.

6.2. The synthetic starting system $\mathcal{S}_0$

To construct the synthetic system $\mathcal{S}_0$, we begin by generating a set of N synthetic cameras, denoted $\{\mathsf{P}^{(0)}_i = \mathsf{K}^{(0)}[\mathsf{R}^{(0)}_i \,|\, \mathbf{t}^{(0)}_i]\}$, where both the intrinsic parameters $\mathsf{K}^{(0)}$ and the extrinsic parameters $(\mathsf{R}^{(0)}_i, \mathbf{t}^{(0)}_i)$ are known. Given the 3D structure of the scene (*i.e.*, the known cylinders), we project them into each synthetic view to compute the corresponding silhouette lines $\mathcal{L}_0 = \{l^{h(0)}_{ij}\}$.

These silhouette lines are then used to construct equations that contain constraints such as (3) and (4). Mimicking what we have done for the orginal system $\mathcal{S}_1$, by collecting all such equations for each visible silhouette in the synthetic setup, we define the system $\mathcal{S}_0(\mathbf{x}, \mathcal{L}_0)$. This system is guaranteed to be consistent by construction, and its solution $\mathbf{x}(0)$ corresponds exactly to the known parameters of the synthetic camera used to generate it.

6.3. Custom silhouette-based homotopy

To solve $\mathcal{S}_1(\mathbf{x}, \mathcal{L}_1)$ we define a custom parametric homotopy H parametrized by line silhouettes. A parametric homotopy [10] is a specialization of the general homotopy framework in which the structure of the system of equations remains fixed, while the parameters vary continuously. In our case, since the 3D cylinders are fixed, all the systems parameters are encoded by the locations in each images of the silhouette lines. The key idea is

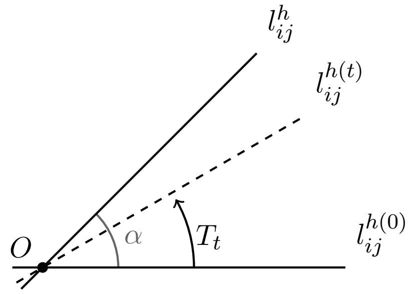


Figure 4: The rotation T_t continuously transforms the synthetic lines $l_{ij}^{h(0)}$ towards the observed lines l_{ij}^h by applying a rotation of an angle $t\alpha$ around the intersection point O. At the end of the homotopy, T_t transforms the synthetic parameters $\mathcal{L}_0$ of our problem to the observed parameters $\mathcal{L}_1$.

to continuously transform over time t the silhouette lines to interpolate between those generated synthetically in $\mathcal{S}_0$ and those observed in the real system $\mathcal{S}_1$.

To this end, we apply to each silhouette line a transformation T_t, as illustrated in Figure 4. Given a synthetic line $l_{ij}^{h(0)} \in \mathcal{L}_0$ and its corresponding observed line $l_{ij}^h \in \mathcal{L}_1$, the transformation T_t rotates the synthetic line around the intersection point $O = l_{ij}^{h(0)} \times l_{ij}^h$ by an angle $\theta_t = t\alpha$, where α is the angle between the two lines. At $t = 0$, the synthetic line remains unchanged; at $t = 1$, it fully aligns with the observed line.

With a slight abuse of notation, we denote by $T_t\mathcal{L}_0$ the set of silhouette lines obtained by applying the transformation T_t to the synthetic lines in $\mathcal{L}_0$. Our silhouette-based parametric homotopy can then be compactly written as:

$$H(\mathbf{x}(t), t) := \mathcal{S}_t(\mathbf{x}(t), T_t\mathcal{L}_0), \tag{9}$$

where $\mathbf{x}(t)$ represents the set of camera parameters being tracked (namely intrinsics and rotations), and $\mathcal{S}_t$ is the system whose structure remains fixed, while the parameters $T_t\mathcal{L}_0$ (the transformed silhouette lines) evolve from synthetic to real.

This, in turn, is solved using Julia homotopy continuation package [2] with an implementation of the ad-hoc homotopy described above.

Convergence to the real solution, while not violating the system constraints, is not always garaunteed. For this reason if the process for a starting *synthetic* system $\mathcal{S}_0$ is halted prematurely by the homotopy package the process is repeated for a new system $\mathcal{S}_0$ by drawing a new set of *synthetic* cameras. For best result cameras should be generated randomly but evenly spaced around the scene.

For the full camera calibration on average 177147 paths (initial solutions) are tracked and 64 possible solutions are found, for the most common skew$= 0$ case 59049 paths are tracked and 112 solutions are found, and finally to localize a calibrated camera 64 paths and 16 solutions. Each solution is substituted in (3). Since the resulting equation in $\mathbf{t}$ is quadratic, we decided to solve it directly using a total degree homotopy. The attained solutions are first filtered based on geometric properties *e.g.*cameras that are not looking at the provided cylinders, and then evaluated using reprojection error to pick the best performing one.

6.4. Handling spurious solutions

Homotopy continuation methods do not allow to force inequality constraints, and as a result, the solver may converge to solutions that are not physically valid, namely a negative focal length or negative skew. To address this issue, we introduce a novel correction step that exploits the fact that the calibration matrix K and the rotation matrix R appear as a product in (7). Specifically, if (K, R) is a solution to (7), then for any invertible matrix C, the pair $(\mathsf{KC}, \mathsf{C}^{-1}\mathsf{R})$ is also a valid solution, since:

$$l_{ij}^{h\top}\mathsf{KCC}^{-1}\mathsf{R}\mathbf{w}_j = l_{ij}^{h\top}\mathsf{KR}\mathbf{w}_j = 0. \tag{10}$$

Based on this observation, we propose to post-correct physically invalid solutions by designing a suitable correction matrix C to enforce positive focal length and skew. A correction is valid if it does not change the absolute values of the intrinsic parameters and if $\mathsf{C}^{-1}\mathsf{R}$ is still a rotation, namely has unitary determinant and is orthogonal:

$$\begin{cases} \det(\mathsf{C}^{-1}\mathsf{R}) = \det(\mathsf{C}^{-1}) = \pm 1 \\ (\mathsf{C}^{-1}\mathsf{R})(\mathsf{C}^{-1}\mathsf{R})^\top = \mathsf{C}^{-1}\mathsf{RR}^\top\mathsf{C}^{-\top} = \mathsf{I}. \end{cases} \tag{11}$$

Note that we allow $\det(\mathsf{C}) = -1$ because a reflection might actually be needed to make the solution geometrically consistent. Using this property, we derive three corrections $\mathsf{C}_{f_x}, \mathsf{C}_{f_y}, \mathsf{C}_{sf_y}$

$$\mathsf{C}_{f_x} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathsf{C}_{f_y} = \begin{pmatrix} 1 & \frac{2\alpha}{f_x} & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathsf{C}_{sf_y} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{12}$$

to correct f_x, f_y and, the skew s and f_y jointly respectively. Note that multiple corrections can be applied to the same solution (K, R) if needed. Unfortunately, we can not find a rotation C_s to correct the skew alone.

7. Experiments

We implemented our solver in Julia and our code is available at *removed for review*. Accuracy and robustness are assessed both in a synthetic controlled scenario and on a real scene.

7.1. Synthetic experiments

We generated 50 different synthetic configurations consisting of $N = 2$ cameras and $M = 3$ cylinders. The cylinders are placed in general 3D positions, while all cameras share the same intrinsic matrix K and are also positioned generically in space, with their optical axes oriented toward the barycenter of the cylinders. We evaluate the performance of our method on the two tasks of camera calibration and camera pose estimation. As regard camera calibration, we consider the relative errors $\Delta f, \Delta c$ and Δs on focal lenghts, principal point and skew. In formulas:

$$\Delta f = \frac{1}{2}\frac{|f_x - \overline{f_x}|}{\overline{f_x}} + \frac{1}{2}\frac{|f_y - \overline{f_y}|}{\overline{f_y}}, \quad \Delta c = \frac{1}{2}\frac{|c_x - \overline{c_x}|}{|\overline{c_x}|} + \frac{1}{2}\frac{|c_y - \overline{c_y}|}{|\overline{c_y}|}, \quad \Delta\alpha = \frac{|\alpha - \overline{\alpha}|}{\overline{\alpha}}. \tag{13}$$

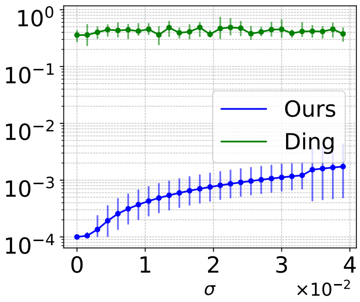
Poses are evaluated in terms of a rotation error in degree $\Delta\mathsf{R}$ and a transaltion error $\Delta\mathbf{t}$:

$$\Delta\mathsf{R} = \cos^{-1}\left(\frac{\mathrm{trace}(\mathsf{R}\overline{\mathsf{R}}^\top) - 1}{2}\right), \quad \Delta\mathbf{t} = \frac{||\mathbf{t} - \bar{\mathbf{t}}||}{||\bar{\mathbf{t}}||} \tag{14}$$

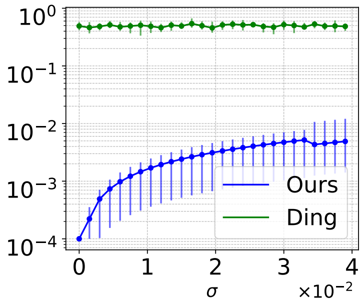
To assess the robustness of our method, we perturbed the observed silhouette lines in $\mathcal{L}_0$ by increasing levels of noise, adopting the same settings as in [4]. Instead of perturbing the line parameters directly, each line is modified by displacing its intersection points with the image boundaries using a random 2D vector whose magnitude matches the prescribed noise level. This results in a more realistic, albeit more severe, form of perturbation.
We compare our camera calibration results with the method proposed by Ding [3]. For a fair comparison, we first assume zero skew ($s = 0$) as the formulation of Ding cannot estimate this parameter. The results are reported in Figures 5a and 5b in logarithmic scale. Our method consistently outperforms Ding's across all noise levels.
We also evaluate separately the estimation of the skew parameter, shown in Figure 5c. While the performance is slightly worse in this case, it is worth noting that, due to the small magnitude of the true skew values, relative error metrics tend to amplify minor fluctuations. In practice, however, for modest noise values the estimated skew remains in the sub 10% range while for higher ones it diverges towards a 50% error.
Results on pose estimation are shown in Figures 5d and 5e. Gummeson [4] represents our closest competitor, as our method can be seen as a generalization of theirs to the case of unknown intrinsics. Our approach consistently outperforms the baseline across all noise levels, demonstrating the advantages of using silhouette-based homotopy continuation over Gröbner basis methods. Finally, Figure 5f reports the success rate of the method. While other approaches that rely on matrix factorizations often fail when the matrices involved become nearly singular, our method consistently finds a physically feasible solution. This highlights the numerical stability of our approach . Note that, as highlighted in Table 1, our method is the only one capable of solving both camera pose and calibration.

7.2. Results on Real Data

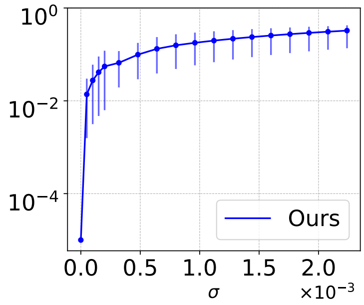
We further evaluated the proposed methods on a real-world scene composed of three cylindrical markers arranged on a desk(Figure 1). Ground truth camera poses were estimated using COLMAP [11, 12]. The 3D position of the cylinders are known.



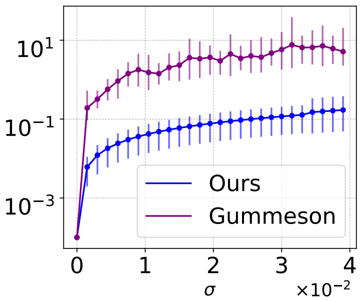
(a) Δf vs. noise.



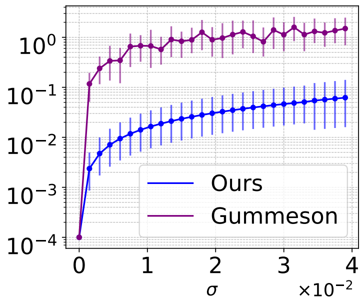
(b) Δc vs. noise.



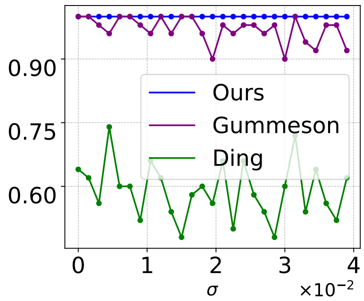
(c) Δs vs. noise.



(d) $\Delta \mathsf{R}$ (degrees) vs. noise.



(e) $\Delta \mathbf{t}$ vs. noise level.



(f) % of successful trials.

Figure 5: Results of our method on the two tasks of camera calibration (top row) and pose estimation (bottom row). In (f) we also report the success rate.

We assume the skew set to zero and use two view of the scene. The estimated intrinsic parameters ($f_x = 1334.72$, $f_y = 1234.91$, $c_x = 848.93$, $c_y = 960$) differ from the COLMAP results ($f_x = 1632.60$, $f_y = 1605.51$, $c_x = 540$, $c_y = 960$) by a mean relative error of 22%, which is still lower than the one achieved by Ding et al. [3] on synthetic data. For localization, $\Delta \mathsf{R}$ is computed on the relative pose of the two cameras and differs from ground truth by 2%. $\Delta \mathbf{t}$ differs by 9%, prior to accounting for symmetries.
For estimating the full camera parameters $(\mathsf{K}, \mathsf{R}, \mathbf{t})$, tracking all 59,049 homotopy paths required 1 minute and 39 seconds. Interestingly, once K is estimated, localization for additional frames requires tracking fewer parameters, resulting in improved speed and accuracy. Execution time drops below one second (single-threaded), with a relative rotation error of 2% and translation error of 0.9%.

8. Conclusions

We presented the first method for both camera calibration and pose estimation from multiple images of known 3D cylinders. By revisiting the geometric constraints of [5], we were able to formulate these two tasks within a unified framework that exploit a novel and custom silouette-based homotopy continuation solver. Experimental results demonstrate superior performance of our approach with respect to existing solutions, both in terms of i) *generality*, ii) *accuracy* and iii) *robustness*. In this work, we focused on the estimation problem under the assumption that the silhouette lines and their associations with 3D cylinders are known. As future work, we plan to address the automatic detection of silhouette lines and their matching to the corresponding cylinders, while also incorporating mechanisms to handle outliers and mismatches in a robust fashion. Currently, the computational complexity of homotopy continuation limits real-time applications. However, future work will explore GPU parallelization to accelerate solution tracking and enable real-time performance.

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A. Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, ...) which supplement the main results contained in the previous sections.

B. Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.

Abstract in lingua italiana

Qui va l'Abstract in lingua italiana della tesi seguito dalla lista di parole chiave.

Parole chiave: qui, le parole chiave, della tesi, in italiano

Acknowledgements

Here you might want to acknowledge someone.